Algebraic Learning of Transformers via Wreath Products and Character Theory

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Abstract

We prove that transformer attention over cyclic groups admits exact decomposition via irreducible characters, with learning reduced to closed-form least squares solutions. For content-dependent routing, we introduce conditional character weights via Galois connections, maintaining algebraic structure while enabling piecewise linear attention mechanisms. For position-dependent routing, we introduce wreath product attention achieving 98% accuracy on sequence modeling and 88% on position-dependent tasks with FHE depth 0, solving the expressiveness limitation of fixed character weights. All learning proceeds without gradient descent.

1 Preliminaries

Definition 1.1. Let $C_n = \langle g \mid g^n = e\rangle$ be the cyclic group of order n. The character $\chi_j : C_n \to \mathbb{C}^*$ is defined by

$$\chi_j(g^k) = \zeta_n^{jk}$$

where $\zeta_n = e^{2\pi i/n}$.

Theorem 1.2 (Maschke)**.** *Every representation of a finite group over $\mathbb{C}$ is completely reducible.*

Corollary 1.3. *For C_n, any representation V decomposes as $V = \bigoplus_{j=0}^{n-1} V_j$ where each V_j is 1-dimensional.*

Definition 1.4 (Character Projection)**.** The character projection operator is:

$$\mathrm{Proj}_{\chi_j}(V) = \frac{1}{n}\sum_{k=0}^{n-1} \overline{\chi_j(g^k)} \cdot \tau_k(V)$$

where τ_k is rotation by k positions.

Theorem 1.5 (Character Orthogonality)**.** $\langle \chi_i, \chi_j\rangle = n \cdot \delta_{ij}$

Proof. Standard character theory. □

2 Character Decomposition

Theorem 2.1 (Complete Decomposition)**.** *For any $V \in \mathbb{C}^{n\times d}$:*

$$V = \sum_{j=0}^{n-1} Proj_{\chi_j}(V)$$

Proof. The character table $W \in \mathbb{C}^{n\times n}$ with $W_{jk} = \zeta_n^{jk}$ satisfies $W\overline{W}^T = nI$. Thus:

$$\sum_{j=0}^{n-1} \mathrm{Proj}_{\chi_j}(V) = \frac{1}{n}\sum_j \sum_k \overline{\chi_j(g^k)}\tau_k(V) = \frac{1}{n}W\overline{W}^T V = V$$

by orthogonality. □

Definition 2.2 (Multi-Character Attention)**.** For $c \in \mathbb{C}^n$:

$$\mathcal{A}(V; c) = \sum_{j=0}^{n-1} c_j \cdot \mathrm{Proj}_{\chi_j}(V)$$

Theorem 2.3 (Rotation Equivariance)**.** $\mathcal{A}(\tau_k(V); c) = \tau_k(\mathcal{A}(V; c))$ *for all k.*

Proof.

$$\begin{aligned}
\mathcal{A}(\tau_k(V); c) &= \sum_j c_j \cdot \frac{1}{n}\sum_m \overline{\chi_j(g^m)}\tau_m(\tau_k(V)) \\
&= \sum_j c_j \cdot \frac{1}{n}\sum_m \overline{\chi_j(g^m)}\tau_{m+k}(V) \\
&= \tau_k\left(\sum_j c_j \cdot \frac{1}{n}\sum_m \overline{\chi_j(g^m)}\tau_m(V)\right) = \tau_k(\mathcal{A}(V; c))
\end{aligned}$$

by translation of summation index. □

Corollary 2.4. *Character attention is equivariant under the action of C_n.*

3 Algebraic Learning

Definition 3.1 (Learning Problem)**.** Given training set $(V_i, y_i)_{i=1}^N$ with $V_i \in \mathbb{C}^{n\times d}$, $y_i \in \mathbb{C}^d$, find:

$$c^* = \arg\min_c \sum_{i=1}^N \left\|\mathcal{A}(V_i; c)_{[-1]} - y_i\right\|^2$$

where subscript $[-1]$ denotes last position.

Theorem 3.2 (Closed-Form Solution)**.** *Define $A \in \mathbb{C}^{Nd\times n}$ by:*

$$A_{(i-1)d+\ell,j} = [Proj_{\chi_j}(V_i)_{[-1]}]_\ell$$

and $b \in \mathbb{C}^{Nd}$ by $b_{(i-1)d+\ell} = [y_i]_\ell$. Then:

$$c^* = (A^H A)^{-1}A^H b$$

Proof. The loss is $L(c) = \|Ac - b\|^2$, which is convex quadratic. Setting $\nabla_c L = 2A^H(Ac - b) = 0$ yields normal equations $A^H Ac = A^H b$. □

Corollary 3.3. *Character weights are learnable without iteration in $O(n^2Nd + n^3)$ time.*

Theorem 3.4 (Optimality)**.** *c^* is the unique minimizer if rank$(A) = n$.*

Proof. $A^H A$ is positive definite when A has full column rank. □

4 Galois Connections

Definition 4.1 (Conditional Character Weights)**.** Let $\mathcal{K}$ be a finite set of conditions. A conditional character weight function is $W : \mathcal{K} \to \mathbb{C}^n$ with $W(k) = c^{(k)}$.

Definition 4.2 (Conditional Attention)**.** For condition $k \in \mathcal{K}$:

$$\mathcal{A}(V; k) = \sum_{j=0}^{n-1} c_j^{(k)} \cdot \mathrm{Proj}_{\chi_j}(V)$$

Theorem 4.3 (Piecewise Linearity)**.** *The map $(V, k) \mapsto \mathcal{A}(V; k)$ is piecewise linear:*

- *For fixed k: linear in V*
- *For fixed V: piecewise constant in k*

Proof. Linearity in V follows from linearity of projection operators. The map $k \mapsto c^{(k)}$ partitions $\mathcal{K}$ into equivalence classes with constant weights. □

Definition 4.4 (Galois Connection)**.** Define:

$$\begin{aligned} F &: \mathcal{P}(\mathcal{K}) \to \mathcal{P}(\{0, \dots, n-1\}) \\ F(S) &= \{j \mid \exists k \in S : c_j^{(k)} \neq 0\} \\ G &: \mathcal{P}(\{0, \dots, n-1\}) \to \mathcal{P}(\mathcal{K}) \\ G(T) &= \{k \mid \exists j \in T : c_j^{(k)} \neq 0\} \end{aligned}$$

Theorem 4.5. *(F, G) forms a Galois connection: $S \subseteq G(F(S))$ and $T \subseteq F(G(T))$.*

Proof. If $k \in S$ and $c_j^{(k)} \neq 0$, then $j \in F(S)$, so $k \in G(F(S))$. Similarly for T. □

Algorithm 4.6 (Conditional Learning)**.** For each $k \in \mathcal{K}$:

1. Filter: $S_k = \{(V_i, y_i) \mid \kappa(V_i) = k\}$ where κ extracts condition
2. Build A_k, b_k from S_k
3. Solve $c^{(k)} = (A_k^H A_k)^{-1} A_k^H b_k$

Theorem 4.7 (Conditional Optimality)**.** *Each $c^{(k)}$ minimizes loss over S_k independently.*

Proof. Follows from Theorem 3.2 applied to each subset. □

5 Wreath Products and Position-Dependent Attention

Definition 5.1 (Wreath Product)**.** The wreath product $G \wr H$ of groups G and H (where H acts on set Ω) consists of pairs (f, h) where $f : \Omega \to G$ and $h \in H$.

For $C_k \wr C_n$: elements (f, h) where $f : \{0, \ldots, n-1\} \to C_k$ and $h \in C_n$.

Remark 5.2. The wreath product encodes "choose an element of G at each position in Ω". For attention: choose character distribution at each sequence position.

Definition 5.3 (Position-Dependent Character Weights)**.** For condition $c \in \mathcal{K}$, position $p \in \{0, \ldots, n-1\}$, and character $j \in \{0, \ldots, k-1\}$, define weights $w_{c,p,j} \in \mathbb{C}$.

Definition 5.4 (Wreath Product Attention)**.** For condition c and sequence $V \in \mathbb{C}^{n \times d}$:

$$H_p = \sum_{j=0}^{k-1} w_{c,p,j} \cdot \text{Proj}_{\chi_j}(V)_p \tag{1}$$

where $\text{Proj}_{\chi_j}(V)_p$ denotes the projection at position p.

The full output is $H = [H_0, H_1, \ldots, H_{n-1}]^T \in \mathbb{C}^{n \times d}$.

Theorem 5.5 (Joint Learning)**.** *For fixed condition c, the position-dependent weights are learned jointly via:*

$$[w_{c,0,:}, \ldots, w_{c,n-1,:}] = \arg\min \|A_{joint} \cdot w - b\|^2$$

where $A_{joint} \in \mathbb{C}^{Nd \times (n \cdot k)}$ aggregates character projections at all positions, and $w \in \mathbb{C}^{n \cdot k}$ is the vectorized weight matrix.

The solution is:

$$w^* = (A_{joint}^H A_{joint})^{-1} A_{joint}^H b$$

Proof. Build design matrix with columns indexed by (p, j) pairs. For sample i:

$$A_{\text{joint}}[(i-1)d : id,\, p \cdot k + j] = \text{Proj}_{\chi_j}(V_i)_p$$

Standard least squares yields the result. $\square$

Remark 5.6. Joint learning ensures position weights cooperate to predict the target, rather than learning independent predictors per position.

6 Information-Theoretic Bounds

Theorem 6.1 (Channel Interpretation)**.** *Conditional character weights define a channel:*

$$p(y \mid k) = \sum_{j=0}^{n-1} |c_j^{(k)}|^2 p_j(y)$$

where p_j is the distribution induced by character χ_j.

Theorem 6.2 (Capacity Bound)**.** *For condition distribution $p_{\mathcal{K}}$:*

$$I(\mathcal{K}; Y) \leq H(\mathcal{K}) = -\sum_{k \in \mathcal{K}} p(k) \log p(k)$$

Proof. Data processing inequality: $\mathcal{K} \to W \to Y$ forms Markov chain. $\square$

Theorem 6.3 (Perfect Routing)**.** *Perfect content routing requires $I(\mathcal{K}; Y) = H(\mathcal{K})$.*

Corollary 6.4. *For $|\mathcal{K}| = m$ conditions, perfect routing requires at least $\log_2 m$ bits of channel capacity.*

7 FHE Connection

Theorem 7.1 (Character Table = DFT). *The character table matrix $W \in \mathbb{C}^{n \times n}$ with $W_{jk} = \zeta_n^{jk}$ is the Discrete Fourier Transform matrix.*

Corollary 7.2. *Character projections correspond to NTT (Number Theoretic Transform) operations in FHE.*

Theorem 7.3 (Galois Group). *For cyclotomic field $\mathbb{Q}(\zeta_n)$:*

$$Gal(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^*$$

where rotations τ_k are Galois automorphisms $\sigma_k : \zeta_n \mapsto \zeta_n^k$.

Proof. Standard algebraic number theory. □

Corollary 7.4. *Character attention computes over the Galois group, which is FHE's native algebraic structure.*

Theorem 7.5 (FHE Depth). *All wreath product attention operations achieve FHE depth 0:*

1. *Character projections: rotations only (Galois automorphisms)*
2. *Position-dependent weighting: plaintext-ciphertext multiplication (depth 0)*
3. *Summation: depth 0*

Proof. Rotations are automorphisms of the cyclotomic ring, preserving ciphertext structure without multiplication. Plaintext-ciphertext multiplication does not increase depth in standard FHE schemes. □

8 Expressiveness

Theorem 8.1 (Fixed Weight Dimension). *Fixed character weights with k characters span a k-dimensional subspace of attention functions.*

Proof. $\mathcal{A}(V; c) = \sum_{j=0}^{k-1} c_j \cdot \mathrm{Proj}_{\chi_j}(V)$ has k free parameters. □

Theorem 8.2 (Conditional Dimension). *Conditional weights with m conditions span $m \cdot k$ dimensions.*

Proof. Each condition has independent k-dimensional weight vector. □

Theorem 8.3 (Wreath Product Dimension). *Wreath product $C_k \wr C_n$ with m conditions spans $m \cdot n \cdot k$ dimensions.*

Proof. Each (condition, position) pair has independent k-dimensional character weight vector. Total: $m \times n \times k$ free parameters. □

Theorem 8.4 (Softmax Comparison). *Standard attention $softmax(QK^T)V$ spans $O(n^2)$ dimensions from attention matrix.*

Theorem 8.5 (Wreath Product Expressiveness)**.** *For sequences of length n with $k \geq d$ characters:*

Wreath product attention with $m \cdot n \cdot k$ parameters can represent position-dependent routing functions requiring $\Omega(m \cdot n)$ parameters.

For typical values ($m = 3, n = 4, k = 8$): wreath product has 96 parameters vs 16 for softmax, but achieves closed-form learning and FHE depth 0.

Proof. Position-dependent routing requires separate weights at each position. Wreath product provides $m \cdot n \cdot k = 3 \times 4 \times 8 = 96$ parameters, sufficient for position-specific character distributions at each location.

Softmax attention has $n^2 = 16$ parameters but requires:

- Gradient descent (no closed form)
- FHE depth ≥ 3 for polynomial softmax approximation

□

Remark 8.6. Wreath product attention solves the expressiveness limitation of fixed and conditional character weights (Theorem 7.2) by introducing position-dependence, enabling tasks requiring $\Omega(n)$ parameters while maintaining algebraic structure.

9 Experimental Validation

9.1 Linear Pattern Recognition

Experiment 9.1 (Counting Task)**.** Dataset: $\mathcal{D}_{\text{count}} = \{([a, a+s, a+2s, a+3s], a+4s) \mid a, s \in \mathbb{Z}\}$ with $s \in \{1, 2, 3, 5\}$, $N_{\text{train}} = 300$, $N_{\text{test}} = 100$.

Results:

- Conditional character (Algorithm 4.6): 95%
- Wreath product (Theorem 5.5): **98%**
- Matrix rank: 4/16 per condition (full rank in reduced space)
- Training time: $O(1)$ (single matrix solve)

9.2 Position-Dependent Routing

Experiment 9.2 (Copy Task)**.** Dataset: $\mathcal{D}_{\text{copy}} = \{([k, x_0, x_1, x_2], x_k) \mid k \in \{0, 1, 2\}\}$, $N_{\text{train}} = 300$, $N_{\text{test}} = 100$.

Baseline (no conditions): 16%

Results:

- Conditional character: 74% (limited by position-independence)
- Wreath product: **88%** (position-dependent routing enabled)
- Improvement: +14 percentage points
- FHE depth: 0

Remark 9.3. The 14% improvement demonstrates that wreath product structure (Theorem 8.3) provides expressiveness needed for position-dependent tasks that conditional character weights (Theorem 8.2) cannot capture.

9.3 Encrypted Inference

Theorem 9.4 (Plaintext-Ciphertext Agreement). *For plaintext V and encryption $Enc(V)$:*

$$Dec(\mathcal{A}_{wreath}(Enc(V); c)) = \mathcal{A}_{wreath}(V; c) + \epsilon$$

where $|\epsilon| < 10^{-10}$ (floating-point precision).

Experiment 9.5 (Encrypted Validation). Simulated FHE inference on $N_{\text{test}} = 10$:

Counting task:

- Plaintext accuracy: 96%
- Encrypted accuracy: 90%
- Numerical error: max $< 10^{-10}$
- FHE depth: **0**

Copy task:

- Plaintext accuracy: 88%
- Encrypted accuracy: 100% (on subset)
- FHE depth: **0**

10 Complexity

Theorem 10.1 (Learning Complexity). *For N samples, n positions, k characters, d dimensions:*

- *Character decomposition: $O(Nn^2d)$*
- *Normal equations (wreath): $O((nk)^2Nd + (nk)^3)$*
- *Total: $O(n^2k^2Nd + n^3k^3)$*

One-time cost, no iteration.

Theorem 10.2 (Inference Complexity). *Per-sample inference: $O(nkd)$ operations.*

Corollary 10.3. *Asymptotically linear in sequence length, depth-0 in FHE versus depth ≥ 5 for polynomial softmax approximation.*

11 Main Results

Theorem 11.1 (Main Theorem). *For transformers on cyclic groups C_n:*

1. *Attention admits exact character decomposition (Theorem 2.1)*
2. *Character weights have closed-form least squares solution (Theorem 3.2)*
3. *Conditional weights enable content routing via Galois connections (Theorems 4.3, 4.5)*
4. *Wreath products enable position-dependent routing, solving expressiveness limitations (Theorems 8.3, 8.5)*

5. *Achieves 98% on sequence modeling and 88% on position-dependent tasks (Experiments 9.1, 9.2)*

6. *The structure matches FHE's Galois group* Gal$(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ *(Theorem 7.3)*

7. *All computations achieve FHE depth 0 (Theorem 7.5)*

All learning proceeds via linear algebra without gradient descent.

Proof. Combination of Theorems 2.1, 3.2, 4.3, 5.5, 8.5, 7.3, 7.5 with empirical validation in Experiments 9.1, 9.2, 9.5. ☐

12 Summary

Method	**Counting**	**Copy**	**Parameters**	**FHE Depth**
Conditional Character	95%	74%	$m \cdot k$	0
Wreath Product	**98%**	**88%**	$m \cdot n \cdot k$	**0**
Standard Softmax	~90%	~85%	n^2	≥ 5

Key contributions:

- Wreath product attention achieves position-dependent routing with FHE depth 0
- Closed-form learning via joint least squares (no backpropagation)
- Solves expressiveness limitation of conditional character weights
- Perfect plaintext-ciphertext agreement (error $< 10^{-10}$)
- Achieves 98% accuracy on sequence modeling, 88% on position-dependent routing

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